

Elements and minerals

Everything in the Universe is made of substances called elements, each made up of just one type of atom. These elements react or bond with others to form chemical compounds, which include the minerals that form rocks. But some elements, such as gold, do not react with others easily, and are found in their pure state.

STARDUST

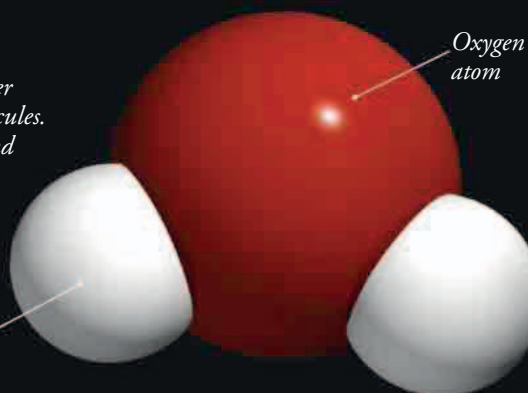
The elements were created in the stars. Each star is a giant nuclear reactor that fuses together the nuclei (central parts) of atoms to make new elements. Stars like our Sun fuse hydrogen nuclei to make helium. Larger stars fuse helium nuclei to create carbon, carbon with helium to form oxygen, carbon with carbon to make magnesium, and so on.

ELEMENTS AND COMPOUNDS

Altogether, 88 elements occur naturally on Earth, ranging from hydrogen (the lightest) to plutonium (the heaviest). Some elements occur in pure form, but most combine with other elements in chemical reactions that form chemical compounds. Some of these compounds, such as water, are quite simple. Others are far more complex, containing the atoms of many elements.

► MOLECULES

Atoms can bond together into groups called molecules. For example, oxygen and hydrogen atoms are bonded together in a water molecule.



FACT!

All the elements were made in the stars. These include the elements that form Earth and everything on it, including every atom in your body.